STEP-UP: AN AUGMENTED REALITY MOBILE APPLICATION FOR PROMOTING PHYSICAL ACTIVITY AMONG UNIVERSITY STUDENTS

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THE PROBLEM AND ITS BACKGROUND

Introduction

Physical activity is an essential component of health and well-being because it supports physical fitness, mental health, disease prevention, and overall quality of life. The World Health Organization (WHO) defines physical activity as any bodily movement produced by skeletal muscles that requires energy expenditure, including movement during leisure time, transportation, work, and daily activities (WHO, 2024). For adults, the WHO recommends at least 150 to 300 minutes of moderate-intensity aerobic physical activity per week, or 75 to 150 minutes of vigorous-intensity aerobic physical activity per week, or an equivalent combination of both (WHO, 2020). These recommendations show that physical activity does not need to be limited to structured exercise, since simple movement such as walking may also contribute to healthier daily routines.

Despite the known benefits of physical activity, physical inactivity remains a continuing global and local concern. The WHO reported that nearly one third of the world's adult population, equivalent to about 1.8 billion adults, did not meet the recommended levels of physical activity in 2022 (WHO, 2024). In the Philippine context, the WHO Physical Activity Profile 2022 showed that physical inactivity remains a concern among adults in the country (WHO, 2022). These findings indicate the need for practical and accessible strategies that can encourage individuals, including students, to become more physically active.

University students may be exposed to sedentary behavior because of academic workload, prolonged sitting, screen time, and limited opportunities or motivation to exercise regularly. Babaeer et al. (2022) emphasized the importance of examining physical activity and sedentary behavior among university students because these behaviors may influence student health and academic-related outcomes. Similarly, Lynch et al. (2022) reported that university students spend considerable time in sedentary activities and that movement-based strategies may help reduce sedentary behavior in higher education settings. These conditions show the need for physical activity interventions that are suitable for the daily routines and interests of university students.

The increasing use of mobile technology provides opportunities to support health-related behaviors in a more accessible and interactive way. Mobile health applications can help users monitor activity, set goals, receive feedback, and view progress. Bi et al. (2024) found that digital health interventions have significant effects on step counts among college students, although their effects on other physical activity outcomes may vary. This suggests that step-based mobile interventions may be useful in encouraging physical activity among students when the application is designed to be engaging, measurable, and easy to use.

Gamification and augmented reality may also help make physical activity more engaging. Gamification uses game-like elements such as points, missions, rewards, progress tracking, avatar customization, and leaderboards to encourage user participation. Mazeas et al. (2022) found that gamified interventions may have positive effects on physical activity, while Xu et al. (2022) reported that mobile health gamification may support participation through features such as goal setting, rewards, points, feedback, and progress monitoring. In addition, augmented reality and location-based interaction can connect digital activities with real-world movement. Lee et al. (2021) explained that Pokémon GO, a location-based augmented reality game, encouraged walking and real-world exploration through game-based interaction.

In response to these concerns, this study focuses on the development and evaluation of STEP-UP: An Augmented Reality Mobile Application for Promoting Physical Activity Among University Students. STEP-UP is designed to encourage walking and physical activity through automatic step tracking, augmented reality walking mode, gamification, rewards, missions, avatar customization, leaderboard ranking, and health guidance features. The application aims to provide students with a more interactive and student-centered way to monitor their movement, view their progress, and become more motivated to engage in step-based physical activity.

The development of STEP-UP addresses the need for a localized digital intervention for university students. Although existing fitness applications provide activity tracking and wellness-related features, many are designed for general users and may not focus on the needs and context of students within a university setting. By combining mobile health, gamification, augmented reality, and step-based monitoring, STEP-UP aims to support physical activity promotion in a way that is accessible, engaging, and appropriate for student users. Therefore, this study seeks to develop and evaluate STEP-UP as a technology-based tool for promoting physical activity among university students.

Project Context

Pampanga State Agricultural University (PSAU) serves as the locale of this study, with its currently enrolled students as the target users of the STEP-UP application. University students often spend long periods attending classes, completing academic requirements, studying, and using digital devices. These routines may contribute to sedentary behavior and limited physical activity. In this context, there is a need for a practical and student-centered approach that can encourage students to become more aware of their daily movement and participate in simple physical activities such as walking.

The project entitled “STEP-UP: An Augmented Reality Mobile Application for Promoting Physical Activity Among University Students” is designed as a mobile application that promotes physical activity through step-based monitoring and interactive features. The application focuses on walking because it is one of the most accessible forms of physical activity for students. It does not require special equipment and can be performed during regular campus routines, such as walking between classrooms, moving around university facilities, or completing daily step goals.

STEP-UP is developed for supported Android smartphones to provide a focused and consistent platform for development, deployment, and evaluation. The Android platform was selected because the application requires mobile functions such as step tracking, location-based services, camera access, and augmented reality-related features. Limiting the application to Android allows the researchers to concentrate the development and testing process on one mobile environment and to better identify device-related limitations during evaluation.

The application is intended to support students in monitoring their daily physical activity. Through automatic step tracking, STEP-UP records the user’s walking activity and allows the user to view progress through activity summaries and step-based records. The system also includes BMI-related information, estimated calories burned, and step recommendations as general wellness references. These features are included to help students become more aware of their activity habits and encourage them to participate in regular movement.

To increase user engagement, STEP-UP integrates gamified features such as points, rewards, missions, leaderboard ranking, and avatar customization. These features are intended to make physical activity more interactive and goal-oriented. Instead of presenting step tracking as a simple record of movement, the application encourages students to complete daily goals and weekly missions while receiving progress-based rewards.

Another important component of STEP-UP is its augmented reality walking mode and location-based features. These features are designed to connect digital interaction with real-world movement. By using augmented reality and map-based functions, the application aims to make walking more engaging for students and provide an interactive experience that supports the purpose of promoting physical activity.

The system also uses cloud-based and local data handling to support user accounts, progress records, rewards, avatar data, and leaderboard information. Online features allow user records to be updated and synchronized, while selected local storage functions help maintain important user data when internet connectivity is unstable or temporarily unavailable. This supports the usability of the application in real campus conditions where internet connection may vary.

In the context of this study, STEP-UP serves as both a developed capstone project and a physical activity promotion tool for university students. The application is not intended to replace professional medical advice, physical education programs, or formal exercise interventions. Instead, it is designed to provide general wellness support, activity awareness, and motivation for students to engage in step-based physical activity. Through this project, the researchers aim to develop and evaluate a localized, student-centered, and technology-based application that responds to the need for more engaging physical activity promotion among university students.

Statement of the Objectives

This study aims to develop and evaluate STEP-UP: An Augmented Reality Mobile Application for Promoting Physical Activity Among University Students. Specifically, the study focuses on the development of a mobile application that integrates step tracking, augmented reality, gamification, and health guidance features to encourage physical activity among university students.

General Objective

The general objective of this study is to develop and evaluate an augmented reality mobile application, STEP-UP, designed to promote physical activity among university students.

Specific Objectives

Specifically, this study aims to:

1. Develop a mobile application that integrates automatic step tracking, augmented reality walking mode, gamification features, BMI-based mission assignment, and health guidance tools.
2. Compare the average daily step counts of the experimental group and control group during the intervention period.
3. Determine whether the STEP-UP application significantly increases physical activity levels among university students.
4. Assess changes in user motivation, autonomy, competence, and behavioral intention toward physical activity after using the application through constructs adapted from Self-Determination Theory (SDT).

5. Evaluate user engagement, enjoyment, immersion, and gameplay experience of the application using selected constructs from the Game Experience Questionnaire (GEQ), as assessed by student users and IT experts.
6. Evaluate the software quality of the STEP-UP application using ISO/IEC 25010 standards in terms of functional suitability, performance efficiency, compatibility, usability, reliability, security, maintainability, and portability.

Significance of the Project

This study is significant because it provides a technology-based approach for promoting physical activity among university students. Since students may experience sedentary routines because of academic workload, prolonged sitting, and frequent use of digital devices, STEP-UP offers an accessible mobile application that encourages walking, progress monitoring, and activity awareness through interactive and gamified features.

Students. The primary beneficiaries of this study are the students of Pampanga State Agricultural University. STEP-UP may help students monitor their daily steps, view their activity progress, complete walking-related missions, and become more aware of their physical activity habits. Through features such as rewards, leaderboard ranking, avatar customization, and health guidance, the application may also encourage students to participate in regular step-based physical activity in a more engaging way.

Pampanga State Agricultural University. The study may benefit the university by providing a localized mobile application that supports student wellness and physical activity promotion. STEP-UP may serve as a technology-based support tool that can encourage students to become more active within the university setting. Since the application is developed specifically for PSAU students, it may also contribute to the university's initiatives related to student health awareness, wellness, and technology-based innovation.

University Infirmary. The study may benefit the University Infirmary by providing a possible support system for promoting student wellness and physical activity awareness. As the intended administrative or supporting office for the application, the Infirmary may use the project as a basis for encouraging students to engage in healthier movement habits. The application may also help support wellness-related activities by providing step-based records, progress monitoring, and general health guidance features.

Physical Education Instructors and Health Advocates. The study may benefit physical education instructors and health advocates by providing a reference for using mobile technology to encourage physical activity among students. STEP-UP may serve as an example of how step tracking, gamification, augmented reality, and health guidance can be integrated to make physical activity more interactive and appealing to university students.

Information Technology Students and Developers. The study may benefit IT students and developers by serving as a reference for developing mobile applications that integrate automatic step tracking, augmented reality, gamification, location-based features, cloud database services, and local data storage. The project may also provide insight into the development and evaluation of Android-based mobile applications designed for health and wellness support.

Future Researchers. The study may benefit future researchers by providing a basis for further studies related to mobile health applications, gamification, augmented reality, and physical activity promotion. The findings of the study may help future researchers improve similar applications, conduct longer testing periods, include other user groups, or explore additional features that may further support student wellness and physical activity.

Scope and Limitations of the Project

This study focuses on the development and evaluation of STEP-UP: An Augmented Reality Mobile Application for Promoting Physical Activity Among University Students. The project is designed for currently enrolled students of Pampanga State Agricultural University as the primary users and respondents of the study. The application aims to promote physical activity through walking-based monitoring, gamification, augmented reality interaction, and general health guidance features.

Scope

STEP-UP includes user registration and login, user profile management, BMI calculation, BMI-based mission assignment, automatic step tracking, activity summary and progress monitoring, local data storage, offline mode with online synchronization, gamification and reward system, daily goals, weekly missions, leaderboard ranking, campus-based map visualization, augmented reality walking mode, avatar customization, health guidance, estimated calories burned, step recommendation, vehicle detection anti-cheat mechanism, and safety reminders.

The study also covers the evaluation of the application based on the objectives of the research. The physical activity level of the respondents will be evaluated by comparing the average daily step counts of the experimental group and control group during the intervention period. The experimental group will use the STEP-UP application, while the control group will continue their usual routine without using the application. This comparison will help determine whether the application contributes to an increase in students' physical activity.

The study will also assess students' motivation toward physical activity using selected constructs adapted from Self-Determination Theory (SDT), including motivation, autonomy, competence, and behavioral intention. The engagement, enjoyment, immersion, and gameplay experience of STEP-UP will be evaluated using selected constructs from the Game Experience Questionnaire (GEQ), as assessed by student users and IT experts. The software quality of the application will be evaluated using ISO/IEC 25010 characteristics, including functional suitability, performance efficiency, compatibility, usability, reliability, security, maintainability, and portability.

The respondents of the study will be selected from the currently enrolled students of Pampanga State Agricultural University. The sample size will be determined using Slovin's Formula based on the official total population of enrolled students. The University Infirmary may serve as the intended administrative or supporting office for the application because the project is related to student wellness and physical activity promotion.

Limitations

The study is limited to Pampanga State Agricultural University students and may not represent students from other universities or institutions. The findings of the study will be based on the selected respondents, the testing period, and the conditions under which the application will be used. Therefore, the results may not fully reflect long-term physical activity behavior or the use of the application in other academic settings.

STEP-UP is limited to supported Android smartphones. The application requires device features such as step tracking support, location services, camera access, and sufficient device capability for augmented reality-related functions. Since Android devices may differ in sensor quality, operating system version, hardware performance, battery optimization settings, and AR compatibility, the performance and accuracy of the application may vary depending on the device used by the respondent.

The step tracking feature is limited to walking and step-based physical activity. It does not measure all forms of physical activity such as swimming, cycling, gym workouts, sports activities, or exercises that do not involve step movement. The recorded step count may also be affected by phone placement, sensor availability, device settings, and user behavior. For this reason, the step count will be treated as an activity indicator rather than a clinical measurement.

The application includes health guidance, BMI information, estimated calories burned, and step recommendations only for general wellness awareness. These features are not intended to provide medical diagnosis, treatment, or professional medical advice. Students with medical conditions, physical limitations, or health concerns should still consult qualified health professionals before engaging in physical activity.

Some features of STEP-UP may require an internet connection, such as account login, data synchronization, leaderboard updates, cloud-based records, and map-related services. Although the application includes local data handling and offline support for selected functions, full system synchronization and online features depend on the availability and stability of the internet connection. Therefore, unstable connectivity may affect the timeliness of updates and the availability of certain online functions.

Definition of Terms

The following terms are defined based on how they are used in this study:

Android. This refers to the mobile operating system used as the platform for the STEP-UP application. In this study, STEP-UP is developed for supported Android smartphones.

Augmented Reality. This refers to a technology that combines digital elements with the real-world environment. In this study, augmented reality is used in the walking mode of STEP-UP to make physical activity more interactive and engaging.

Automatic Step Tracking. This refers to the feature of STEP-UP that records the user's walking activity through the supported step tracking capability of the Android device. In this study, step tracking is used to monitor daily steps and support activity summaries, goals, missions, rewards, and leaderboard ranking.

Body Mass Index. This refers to a general body measurement based on height and weight. In this study, BMI is used only as a general wellness reference and as part of the application's health guidance and step recommendation features.

Control Group. This refers to the group of student respondents who will not use the STEP-UP application during the intervention period. In this study, the control group will continue their usual routine and will be compared with the experimental group.

Experimental Group. This refers to the group of student respondents who will use the STEP-UP application during the intervention period. In this study, the experimental group will be evaluated to determine whether the application contributes to changes in physical activity.

Firebase. This refers to the cloud-based platform used to support selected backend functions of STEP-UP. In this study, Firebase is used for user authentication, data storage, synchronization, user records, rewards, and leaderboard-related information.

Game Experience Questionnaire. This refers to an evaluation tool used to assess user experience in game-based or gamified systems. In this study, selected GEQ constructs are used to evaluate engagement, enjoyment, immersion, and gameplay experience of STEP-UP as assessed by student users and IT experts.

Gamification. This refers to the use of game-like elements in a non-game system. In this study, gamification includes missions, rewards, points, leaderboard ranking, progress monitoring, and avatar customization to encourage student participation in physical activity.

Health Guidance. This refers to the general wellness information provided by the STEP-UP application. In this study, health guidance includes physical activity reminders, BMI-related information, estimated calories burned, step recommendations, and safety reminders, but it does not provide medical diagnosis, treatment, or professional medical advice.

ISO/IEC 25010. This refers to a software quality model used to evaluate the quality of a software product. In this study, ISO/IEC 25010 is used to evaluate STEP-UP in terms of functional suitability, performance efficiency, compatibility, usability, reliability, security, maintainability, and portability.

Mapbox. This refers to the map and location-based service used in the STEP-UP application. In this study, Mapbox supports campus-based map visualization and location-related features.

Physical Activity. This refers to bodily movement that requires energy expenditure. In this study, physical activity focuses mainly on walking and step-based movement recorded through the STEP-UP application.

Self-Determination Theory. This refers to a theory of motivation that explains how autonomy, competence, and motivation influence behavior. In this study, SDT is used to assess changes in students' motivation, autonomy, competence, and behavioral intention toward physical activity.

Vehicle Detection Anti-Cheat Mechanism. This refers to the feature of STEP-UP that helps prevent inaccurate step records caused by movement that may not represent walking. In this study, the mechanism is used to reduce possible cheating or inaccurate activity recording when the detected movement exceeds the expected walking condition.

BMI-Based Mission Assignment. This refers to the feature of STEP-UP that assigns activity-related missions based on the user's BMI result. In this study, BMI-based mission assignment is used to provide step-related goals or missions as general wellness support and not as medical advice or clinical prescription.

STEP-UP. This refers to the augmented reality mobile application developed in this study. STEP-UP is designed to promote physical activity among university students through step tracking, gamification, augmented reality walking mode, rewards, missions, avatar customization, leaderboard ranking, and health guidance features.

Review of Related Literature and Systems

This chapter presents the literature, studies, frameworks, and related systems that support the development and evaluation of STEP-UP: An Augmented Reality Mobile Application for Promoting Physical Activity Among University Students. The review focuses on physical inactivity and sedentary behavior among university students, physical activity promotion, mobile health applications, smartphone-based step tracking, gamification, augmented reality, location-based technologies, and software quality evaluation. These topics are included because they provide the foundation for understanding the need, design, and evaluation of STEP-UP.

The reviewed literature also explains the basis of the major features of the application, including automatic step tracking, BMI-based mission assignment, rewards, leaderboard ranking, avatar customization, augmented reality walking mode, campus-based map visualization, and vehicle detection anti-cheat mechanism. Since STEP-UP is designed as an Android-based mobile application, the review also discusses the relevance of mobile devices, sensors, location services, and Android-supported features in physical activity monitoring. These discussions help establish why the application was developed and how its features are connected to previous studies and existing technologies.

This chapter further includes the theoretical and evaluation frameworks used in the study. Self-Determination Theory (SDT) serves as the basis for evaluating motivation, autonomy, competence, and behavioral intention toward physical activity. The Game Experience Questionnaire (GEQ) supports the evaluation of engagement, enjoyment, immersion, and gameplay experience, while ISO/IEC 25010 serves as the basis for evaluating the software quality of the STEP-UP application. Related systems are also reviewed to identify existing applications with similar features and to explain how STEP-UP differs as a localized and student-centered physical activity application.

Physical Inactivity and Sedentary Behavior Among University Students

Physical inactivity and sedentary behavior remain important health concerns because regular physical activity contributes to physical health, mental well-being, disease prevention, and overall wellness. The World Health Organization (WHO) defines physical activity as any bodily movement produced by skeletal muscles that requires energy expenditure, including movement during leisure time, transportation, work, and daily activities (WHO, 2024). The WHO also recommends that adults perform at least 150 to 300 minutes of moderate-intensity aerobic physical activity per week, or 75 to 150 minutes of vigorous-intensity aerobic physical activity per week, or an equivalent combination of both (WHO, 2020). These recommendations show that physical activity does not have to be limited to structured exercise, since regular movement such as walking may also contribute to healthier lifestyle habits.

Despite these recommendations, physical inactivity continues to be a global and local concern. The WHO reported that 31% of adults worldwide, equivalent to about 1.8 billion adults, did not meet the recommended levels of physical activity in 2022 (WHO, 2024). In the Philippine context, the WHO Physical Activity Philippines 2022 Country Profile provides a country-level reference for physical activity monitoring and promotion (WHO, 2022). These findings show the need for practical and accessible health promotion strategies that encourage individuals, including students, to engage in regular movement-based activities.

University students may be vulnerable to sedentary behavior because academic routines often involve prolonged sitting, studying, completing requirements, attending classes, and using digital devices. Babaeer et al. (2022) examined the relationship of physical activity and sedentary behavior with educational outcomes among university students and emphasized the importance of understanding these behaviors in the higher education setting. Lynch et al. (2022) also reported that university students are mostly sedentary in tertiary education settings and that classroom movement breaks and physically active learning may help reduce sedentary behavior, fatigue, and improve well-being. These studies suggest that physical activity promotion among university students should consider their academic routines and daily movement opportunities.

Walking is one of the simplest and most accessible forms of physical activity for students because it can be performed during regular daily routines without requiring special equipment. The WHO identifies walking as one of the common ways to be physically active, along with cycling, sports, active recreation, and play (WHO, 2024). In a university setting, walking may occur when students move between classrooms, visit campus facilities, or participate in step-based activities. This makes walking a practical focus for technology-based interventions that aim to increase students' physical activity.

These findings establish the need for STEP-UP as a mobile-based intervention that encourages walking and reduces sedentary habits among university students. Since the application focuses on step tracking, walking goals, and movement-based missions, it responds to the need for a practical and measurable approach to physical activity promotion. The reviewed literature supports the development of STEP-UP as an accessible, student-centered, and technology-based tool for encouraging students to become more aware of their daily physical activity.

Physical Activity Promotion and Student Wellness

Physical activity promotion is important in supporting student wellness because regular movement contributes to physical health, mental well-being, and overall quality of life. The World Health Organization (WHO) emphasized that physical activity provides significant physical and mental health benefits and may help reduce symptoms of depression and anxiety, improve brain health, and support overall well-being (WHO, 2024). These benefits are relevant to university students because academic workload, prolonged sitting, and frequent screen use may affect their physical and mental wellness.

Physical activity does not have to be limited to sports, gym workouts, or structured exercise programs. The WHO stated that common ways to be active include walking, cycling, sports, active recreation, and play, and that any amount of physical activity is better than none (WHO, 2024). This is important in the university setting because walking is one of the most accessible forms of physical activity that students can perform during their daily routines, such as moving between classrooms, walking around campus, or completing simple step-based goals.

Studies also show that physical activity interventions can support student wellness in higher education. Donnelly et al. (2024) reviewed studies that used physical activity to improve mental health and quality of life among higher education students. Their review supports the value of physical activity interventions as a possible approach for improving student wellness. Similarly, Lynch et al. (2022) reported that classroom movement breaks and physically active learning are feasible in university settings and may help reduce sedentary behavior and fatigue while improving focus. These findings suggest that movement-based strategies can be useful in encouraging healthier routines among students.

The reviewed literature supports the purpose of STEP-UP in promoting walking as a simple and accessible form of physical activity. By using step tracking, progress monitoring, missions, rewards, and health guidance, the application aims to encourage students to include more movement in their daily routines. This makes STEP-UP relevant as a student-centered tool for physical activity promotion and wellness support within the university setting.

Mobile Health Applications and Digital Physical Activity Interventions

Mobile health applications have become useful tools for supporting health-related behaviors because they are accessible, portable, and commonly used through smartphones. The World Health Organization (WHO, 2021) recognized digital health as an important approach for improving health services and supporting health promotion through digital technologies. In the context of physical activity, mobile applications can provide activity monitoring, goal setting, reminders, feedback, and progress tracking, which may help users become more aware of their movement habits.

Smartphones are suitable for physical activity interventions because they contain built-in features such as sensors, internet connectivity, and location services. These features allow applications to monitor steps, record user progress, and provide feedback based on user activity. Romeo et al. (2019) found that smartphone applications may help increase physical activity, although the effectiveness of these applications depends on their design and the level of user engagement.

Digital health interventions have also been studied among college students. Bi et al. (2024) reported that digital health interventions can significantly increase step counts among college students, showing that mobile-based approaches may be useful in encouraging physical activity in higher education settings. However, the study also suggests that not all physical activity outcomes improve equally, which means that application design and user motivation remain important considerations.

These findings support the development of STEP-UP as a mobile health application for promoting physical activity among university students. Instead of relying only on activity monitoring, STEP-UP integrates automatic step tracking, gamification, augmented reality walking mode, rewards, leaderboard ranking, and health guidance. These features aim to make physical activity monitoring more engaging and suitable for student users.

Smartphone-Based Step Tracking and Activity Monitoring

Smartphone-based step tracking is a practical method for monitoring physical activity because many students already use smartphones in their daily routines. Modern smartphones contain built-in sensors and mobile services that can support activity monitoring, including motion sensing and location-related functions. Android devices may also support a step counter sensor, which reports the number of steps taken by the user while the sensor is activated (Android Developers, 2026). These features make smartphones useful tools for recording walking activity and supporting physical activity awareness.

Step count is commonly used in physical activity monitoring because it is simple, measurable, and easy for users to understand. Paluch et al. (2022) found that higher daily step counts were associated with lower all-cause mortality among adults, although the study also noted that there are no universal public health guidelines that prescribe one exact number of steps per day for all individuals. This shows that step count can be used as a useful indicator of physical activity, but it should be interpreted as a general activity measure rather than a clinical requirement.

However, smartphone-based step tracking also has limitations. Straczkiewicz et al. (2021) explained that smartphones can be used for human activity recognition and health research, but data quality may be affected by factors such as phone placement, device orientation, missing data, and sensor-related issues. Similarly, Parmenter et al. (2022) emphasized that smartphone approaches for measuring physical activity among young people still require attention to validity and reliability. These findings show that step tracking is useful for monitoring walking activity, but its accuracy may vary depending on the device, sensor quality, phone placement, and user behavior.

Since STEP-UP uses step count as one of its main activity indicators, the application also includes safeguards to help reduce inaccurate activity records. Smartphone and location-related data may be used to identify movement conditions that are not consistent with walking behavior. Android location services can provide speed information in meters per second, and smartphone-based activity recognition studies have used device movement data to distinguish between motorized and non-motorized movement patterns (Android Developers, 2026; Straczkiewicz et al., 2021). This supports the inclusion of a vehicle detection anti-cheat mechanism in STEP-UP to help limit step recording when detected movement exceeds expected walking conditions.

These studies justify the use of step count as a measurable activity indicator in STEP-UP. Through smartphone-based step tracking, the application can record walking activity, support daily goals and weekly missions, estimate progress, and provide data for rewards and leaderboard ranking. At the same time, the reviewed literature supports the need to treat step count as an activity indicator rather than a perfect measurement, especially because smartphone-based tracking may be affected by device and user-related factors.

Gamification and Motivation in Fitness Applications

Gamification refers to the use of game-like elements in non-game settings to improve user motivation, engagement, and participation. In mobile health and fitness applications, gamification may include goals, points, rewards, badges, missions, progress bars, leaderboards, feedback, and avatars. These features are used to make physical activity more interactive and goal-oriented, especially for users who may lack motivation to exercise regularly.

Gamification has been studied as a strategy for encouraging physical activity through mobile health applications. Xu et al. (2022) reported that mobile health gamification interventions may support participation in physical activity through features such as goal setting, rewards, points, feedback, and progress monitoring. Mazeas et al. (2022) also found that gamified interventions may have positive effects on physical activity, while Yang et al. (2022) reported that gamified smartphone applications can produce improvements in physical activity outcomes. These findings suggest that gamification can help make physical activity more engaging when game-like features are properly integrated into the application.

In STEP-UP, gamification is applied through missions, rewards, points, avatar customization, leaderboard ranking, and progress monitoring. The application also includes BMI-based mission assignment, where the user's BMI result is used as a general wellness reference for assigning step-related missions. The World Health Organization recommends regular aerobic physical activity for adults, while the Centers for Disease Control and Prevention provides adult BMI categories such as underweight, healthy weight, overweight, and obesity (WHO, 2020; CDC, 2024). Since STEP-UP mainly promotes walking, these general health references are translated into step-based missions within the application.

Step-count literature also supports the use of steps as a measurable indicator of physical activity. Paluch et al. (2022) found that higher daily step counts were associated with lower all-cause mortality among adults, although the study did not establish one universal step goal for all individuals. For this reason, the specific missions and step targets in STEP-UP are treated as application-based design decisions and general wellness references rather than medical prescriptions. The BMI-based missions and step recommendations should also be reviewed by medical and fitness experts to ensure that they are appropriate for university student users.

The literature supports the integration of gamified missions and rewards in STEP-UP because these features may encourage students to participate in walking-based activities and monitor their progress. By combining step tracking with missions, points, rewards, avatar customization, and leaderboard ranking, STEP-UP aims to make physical activity more engaging and motivating for university students. This approach also allows the application to transform walking from a routine activity into a more interactive and goal-oriented experience.

Augmented Reality and Location-Based Technologies in Physical Activity

Augmented reality is a technology that combines virtual elements with the real-world environment, allowing users to interact with digital content while viewing or moving within physical spaces. Azuma (1997) described augmented reality as a system that combines real and virtual content, supports real-time interaction, and aligns virtual objects with the real environment. In mobile applications, augmented reality can be supported by smartphone cameras, sensors, and location-related services to create interactive experiences connected to the user’s real-world surroundings.

Location-based technologies are also useful in physical activity applications because they allow mobile systems to respond to the user’s real-world movement and position. Through map visualization and location-related features, mobile applications can encourage users to walk, explore, and complete activity-based tasks in actual environments. This makes location-based technology relevant to STEP-UP because the application aims to connect digital interaction with walking and movement in the university setting.

Several studies have shown the potential of augmented reality and location-based applications in encouraging physical activity. Pokémon GO is commonly used as an example because it combines location-based gameplay, mobile interaction, and real-world exploration. Lee et al. (2021) reviewed studies on Pokémon GO and reported that the game may promote physical activity, especially walking, while also producing psychological and social outcomes. Similarly, Khamzina et al. (2020) found that Pokémon GO was associated with short-term increases in physical activity, although long-term effects were less consistent.

These findings support the inclusion of augmented reality walking mode and location-based features in STEP-UP. By using AR interaction and map-based visualization, STEP-UP aims to make walking more engaging for university students. The application does not only record steps but also provides an interactive experience through missions, rewards, avatar customization, leaderboard ranking, and AR walking mode. This supports the goal of transforming walking into a more engaging and student-centered physical activity experience.

Theoretical and Evaluation Frameworks Related to the Study

The development and evaluation of STEP-UP are supported by theoretical and evaluation frameworks that are related to motivation, game experience, and software quality. Since the application is designed to promote physical activity through step tracking, gamification, augmented reality, missions, rewards, and health guidance, it requires evaluation not only in terms of technical performance but also in terms of user motivation and experience. For this reason, this study uses Self-Determination Theory (SDT), the Game Experience Questionnaire (GEQ), and ISO/IEC 25010 as the main bases for evaluating the application.

Self-Determination Theory is a theory of human motivation that explains how people become motivated when their psychological needs are supported. Deci and Ryan (2000) explained that motivation is influenced by the satisfaction of basic psychological needs, particularly autonomy, competence, and relatedness. In this study, SDT serves as the basis for assessing students' motivation, autonomy, competence, and behavioral intention toward physical activity. These constructs are relevant because STEP-UP encourages users to make progress, complete missions, monitor their activity, and become more aware of their physical activity behavior.

SDT is also relevant to digital behavior change technologies because applications designed to influence behavior often depend on user motivation and continued participation. Alberts et al. (2024) discussed the use of SDT in behavior change technologies and emphasized that technology should support meaningful and sustained motivation rather than only short-term engagement with the application. This supports the evaluation of STEP-UP because the application is not only intended to entertain users, but also to encourage students to value physical activity and continue engaging in walking-based movement.

The Game Experience Questionnaire is used to assess user experience in game-based or gamified systems. IJsselsteijn et al. (2013) explained that the GEQ has a modular structure and that its core questionnaire assesses components such as immersion, flow, competence, affect, tension, and challenge. In this study, selected GEQ constructs are adapted to evaluate the engagement, enjoyment, immersion, and gameplay experience of STEP-UP. This is appropriate because the application includes game-like features such as missions, rewards, leaderboard ranking, avatar customization, and augmented reality walking mode.

The GEQ is relevant not only for student users but also for IT experts because STEP-UP includes interactive and gamified functions that should be evaluated from both user experience and technical perspectives. Student users can assess whether the application is enjoyable, engaging, and immersive during actual use, while IT experts can assess whether the game-like features are properly designed, understandable, and appropriate for the purpose of the application. This supports the adviser's recommendation that GEQ-related evaluation should be answered by both students and IT experts.

ISO/IEC 25010 is used as the basis for evaluating the software quality of STEP-UP. The ISO/IEC 25010:2011 product quality model includes eight characteristics for software product quality: functional suitability, performance efficiency, compatibility, usability, reliability, security, maintainability, and portability (ISO/IEC, 2011). These characteristics are appropriate for evaluating STEP-UP because the application must function correctly, perform efficiently, work on supported Android devices, provide a usable interface, protect user-related data, and allow future improvement or maintenance.

Together, these frameworks provide a more complete basis for evaluating STEP-UP. SDT focuses on students' motivation toward physical activity, GEQ focuses on the experience of using the gamified and augmented reality features, and ISO/IEC 25010 focuses on the technical quality of the application. By using these frameworks, the study can evaluate STEP-UP not only as a mobile application but also as a student-centered physical activity promotion tool.

Related Systems

Several existing mobile applications and systems provide features related to physical activity tracking, fitness monitoring, gamification, and location-based interaction. These systems are reviewed to identify common features that may support physical activity promotion and to determine how STEP-UP differs from existing applications. The related systems include general fitness applications, step tracking platforms, gamified fitness applications, and location-based augmented reality games.

Google Fit is a health and fitness application that allows users to track activities and view progress toward their fitness goals through an Android device or supported wearable device (Google Fit Help, n.d.). Samsung Health also provides health and wellness tracking features, including activity, sleep, step count, and other wellness-related records; its step tracking uses movement data from supported Samsung Galaxy and Android devices (Samsung, 2026). Similarly, the Fitbit application supports step and distance tracking through a smartphone or paired fitness tracker and allows users to set goals, monitor progress, and view activity records (Google, 2023). These systems show how mobile applications can support physical activity awareness through step tracking, goals, and progress monitoring.

Strava is another fitness application that focuses on activity recording, progress monitoring, and goal setting. Strava allows users to set goals based on distance, time, or activity count and provides tools for tracking progress over time (Strava Support, n.d.). This system shows the importance of measurable goals in fitness applications. However, fitness platforms such as Google Fit, Samsung Health, Fitbit, and Strava are generally designed for broad fitness monitoring and activity tracking. They provide useful references for STEP-UP, but STEP-UP is developed specifically for university students and focuses on walking, campus-based movement, gamified missions, and student wellness support.

Location-based and gamified applications also provide important references for STEP-UP. Pokémon GO is a location-based mobile application that connects the real world with virtual content and encourages users to explore physical locations (The Pokémon Company, 2023). Studies on Pokémon GO show that location-based gameplay may encourage walking and physical activity, although its long-term effects may vary (Khamzina et al., 2020; Lee et al., 2021). Zombies, Run! is another gamified fitness application that turns walking, jogging, or running into story-based missions, showing how narrative and mission-based design can make physical activity more engaging (Zombies, Run! Support, 2026).

The reviewed systems show that step tracking, progress monitoring, goals, rewards, missions, location-based interaction, and gamified experiences are useful features in physical activity applications. However, STEP-UP differs from these systems because it combines automatic step tracking, BMI-based mission assignment, augmented reality walking mode, campus-based map visualization, rewards, leaderboard ranking, avatar customization, health guidance, and a vehicle detection anti-cheat mechanism in one localized application for university students. This makes STEP-UP more focused on the needs and context of PSAU students rather than general fitness users.

These related systems provide design references for STEP-UP, but the proposed application is not intended to copy existing platforms. Instead, STEP-UP applies selected concepts from physical activity tracking, gamification, and location-based interaction to create a student-centered mobile application. The review of related systems supports the development of STEP-UP as a localized and interactive tool for promoting walking-based physical activity among university students.

Summary of Key Findings of Previous Research

The reviewed literature shows that physical inactivity and sedentary behavior remain important concerns, especially among adults and university students. The World Health Organization emphasized that regular physical activity supports physical health, mental well-being, and disease prevention, while insufficient physical activity remains a global concern (WHO, 2024). Studies on university students also show that sedentary routines may be influenced by academic workload, prolonged sitting, and learning-related activities (Babaeer et al., 2022; Lynch et al., 2022). These findings support the need for interventions that can encourage students to become more physically active within their regular routines.

Previous studies also show that mobile health applications and digital interventions can support physical activity promotion. Smartphone-based applications may help users monitor activity, receive feedback, set goals, and view progress over time (Ferguson et al., 2021). Digital health interventions among college students have also shown significant effects on step counts, although results may vary depending on the design and implementation of the intervention (Bi et al., 2024). These findings suggest that mobile applications can be useful tools for promoting physical activity when they provide measurable, accessible, and engaging features.

Step tracking was also identified as an important feature in physical activity applications because step count is simple, measurable, and easy for users to understand. Step-count literature shows that higher daily step counts are associated with health benefits among adults, but it also emphasizes that there is no single universal step goal that applies to all users (Paluch et al., 2022). Smartphone-based activity monitoring may also be affected by device placement, sensor quality, missing data, and user behavior (Straczkiewicz et al., 2021; Parmenter et al., 2022). These findings support the use of step count in STEP-UP while also justifying the need for safeguards, such as the vehicle detection anti-cheat mechanism, to help reduce inaccurate activity records.

The literature further shows that gamification can help make physical activity more engaging. Features such as goals, rewards, points, feedback, missions, progress monitoring, leaderboards, and avatars may encourage users to participate and continue using mobile health applications (Xu et al., 2022; Mazeas et al., 2022; Yang et al., 2022). For STEP-UP, these findings support the inclusion of gamified missions, rewards, avatar customization, and leaderboard ranking. The BMI-based mission assignment feature is also supported by general physical activity recommendations and BMI categories, but the exact step targets and mission values are treated as application-based design decisions that require review by medical and fitness experts (WHO, 2020; CDC, 2024; Paluch et al., 2022).

Studies on augmented reality and location-based applications also show that real-world interaction can encourage walking and exploration. Research on Pokémon GO indicates that location-based gameplay may increase physical activity, especially walking, although long-term effects may vary (Khamzina et al., 2020; Lee et al., 2021). Related systems such as Google Fit, Samsung Health, Fitbit, Strava, Pokémon GO, and Zombies, Run! also show that step tracking, goals, progress monitoring, missions, and location-based interaction are common features in existing fitness and activity applications. However, these systems are generally designed for broad users rather than a specific university setting.

Taken together, the previous studies and related systems provide a strong basis for the development of STEP-UP. The findings support the integration of step tracking, gamification, BMI-based missions, augmented reality walking mode, map-based interaction, and software quality evaluation. At the same time, the reviewed literature shows the importance of evaluating STEP-UP not only as a technical application but also as a student-centered tool for promoting physical activity among university students.

Discussion of the Literature Review's Implications for the Current Study

The reviewed literature shows that physical inactivity and sedentary behavior among university students require practical and accessible interventions. Since students may experience prolonged sitting, academic workload, and frequent use of digital devices, physical activity promotion should be designed in a way that fits their daily routines. The findings from WHO (2024), Babaeer et al. (2022), and Lynch et al. (2022) support the need for a student-centered approach that encourages movement without requiring complicated equipment or highly structured exercise programs.

The literature also supports the use of mobile health applications as tools for physical activity promotion. Studies on digital health interventions and smartphone-based activity tracking show that mobile applications can help users monitor activity, set goals, receive feedback, and view progress (Ferguson et al., 2021; Bi et al., 2024). This provides a basis for STEP-UP's automatic step tracking, activity summary, progress monitoring, daily goals, and weekly missions. Since walking is accessible and step count is easy to understand, STEP-UP uses step-based monitoring as its main activity indicator.

The use of gamification in STEP-UP is also supported by previous studies. Gamified interventions have been shown to support physical activity through elements such as goals, rewards, points, feedback, and progress monitoring (Xu et al., 2022; Mazeas et al., 2022; Yang et al., 2022). These findings justify the inclusion of missions, rewards, leaderboard ranking, avatar customization, and progress-based feedback in STEP-UP. The BMI-based mission assignment feature is also connected to general physical activity recommendations and BMI categories, but the exact mission values are treated as application-based design decisions that require validation from medical and fitness experts (WHO, 2020; CDC, 2024; Paluch et al., 2022).

The reviewed studies further support the use of augmented reality and location-based technologies to make walking more interactive. Research on Pokémon GO shows that location-based gameplay may encourage walking and real-world exploration, although long-term effects may vary (Khamzina et al., 2020; Lee et al., 2021). This supports the inclusion of augmented reality walking mode and campus-based map visualization in STEP-UP. These features allow the application to connect digital interaction with actual movement within the university environment.

The limitations discussed in previous studies also influenced the design of STEP-UP. Smartphone-based physical activity tracking may be affected by phone placement, sensor quality, device settings, and user behavior (Straczkiewicz et al., 2021; Parmenter et al., 2022). Because of this, STEP-UP treats step count as a general activity indicator rather than a clinical measurement. The application also includes a vehicle detection anti-cheat mechanism to help reduce inaccurate step records when detected movement exceeds expected walking conditions.

The evaluation frameworks reviewed in this chapter provide the basis for assessing STEP-UP. Self-Determination Theory supports the assessment of motivation, autonomy, competence, and behavioral intention toward physical activity. The Game Experience Questionnaire supports the evaluation of engagement, enjoyment, immersion, and gameplay experience among student users and IT experts. ISO/IEC 25010 provides the basis for evaluating the software quality of STEP-UP in terms of functional suitability, performance efficiency, compatibility, usability, reliability, security, maintainability, and portability.

Based on the reviewed literature, the development and evaluation of STEP-UP are supported as a localized, Android-based, and student-centered mobile application for promoting physical activity. The reviewed studies justify the major features of the system, including step tracking, gamification, BMI-based missions, augmented reality walking mode, campus-based map visualization, health guidance, and anti-cheat support. These findings also show that STEP-UP should be evaluated not only based on technical quality but also based on its ability to support student motivation, engagement, and physical activity awareness.

Technical Background

This chapter presents the technical background of STEP-UP: An Augmented Reality Mobile Application for Promoting Physical Activity Among University Students. It discusses the design, tools, hardware requirements, system architecture, network design, and major functions used in the development of the application. Since STEP-UP is an Android-based mobile application, this chapter explains how the system was developed using mobile technologies, device sensors, augmented reality, location-based services, gamification, cloud services, and local data handling.

The chapter also describes how the different components of the application work together to support the purpose of the study. These components include user registration and login, BMI calculation, BMI-based mission assignment, automatic step tracking, activity progress monitoring, daily goals, weekly missions, rewards, leaderboard ranking, avatar customization, campus-based map visualization, augmented reality walking mode, health guidance, safety reminders, offline support, online synchronization, and vehicle detection anti-cheat mechanism.

This chapter further presents the software tools and hardware specifications needed to run the application. It explains how Unity, C#, Android-supported device features, Firebase, Mapbox, ARCore-related functions, and local storage were used in the system. The chapter also includes the architectural design, network design, and project management approach followed during the development of STEP-UP.

Through this chapter, the technical foundation of STEP-UP is explained in terms of how the application was planned, developed, organized, and prepared for testing. The discussion focuses on the actual implementation of the system and the technologies used to support the application's physical activity promotion features.

Design Specification

The design specification of STEP-UP presents the planned structure, user flow, and functional organization of the application. STEP-UP was designed as an Android-based mobile application that promotes physical activity among university students through walking-based monitoring, gamified activities, augmented reality interaction, map visualization, and health guidance. The design focused on making the application easy to use, interactive, and appropriate for student users.

The application was organized into different modules to make the development process more manageable. These modules include user registration and login, user profile management, BMI calculation, BMI-based mission assignment, automatic step tracking, activity progress monitoring, daily goals, weekly missions, rewards, leaderboard ranking, avatar customization, augmented reality walking mode, campus-based map visualization, health guidance, safety reminders, local data storage, online synchronization, and vehicle detection anti-cheat mechanism. Each module performs a specific function but works together as part of one system.

The user flow of STEP-UP begins with registration or login. After accessing the application, the user provides the required profile information, including height and weight. These data are used by the system to calculate the user's Body Mass Index (BMI). The BMI result is used only as a general wellness reference and serves as the basis for assigning step-related missions. After this process, the user can access the main interface, view daily progress, complete assigned missions, use the AR walking mode, view the campus map, customize an avatar, and check rewards or leaderboard ranking.

The step tracking design focuses on recording the user's walking activity through the supported step tracking capability of the Android device. The recorded steps are used by the application to update activity progress, determine goal completion, provide rewards, and support leaderboard ranking. Since step tracking may be affected by device conditions and user behavior, the application also includes a vehicle detection anti-cheat mechanism to help reduce inaccurate step records caused by movement that may not represent walking.

The gamification design of STEP-UP was included to make physical activity more engaging for students. Instead of presenting step tracking only as a record of movement, the application uses missions, points, rewards, leaderboard ranking, and avatar customization to encourage users to participate in walking-based activities. The daily goals and weekly missions are designed to give users clear activity targets, while the reward and avatar systems provide additional motivation for continued use.

The augmented reality and map design of STEP-UP connects digital interaction with real-world movement. The AR walking mode allows users to experience physical activity in a more interactive way using the device camera and AR-supported functions. The campus-based map visualization supports location-related interaction by helping users connect the application experience with the university environment.

The data design of STEP-UP includes both local storage and online synchronization. Local data storage is used to support selected functions when internet connectivity is unstable or temporarily unavailable. Online synchronization is used to update account-related records, progress data, rewards, and leaderboard information when an internet connection is available. This design helps the application remain usable in real campus conditions where internet access may vary.

This section shows that STEP-UP was designed as a modular and student-centered mobile application. Its design combines step tracking, BMI-based mission assignment, gamification, augmented reality, map visualization, local storage, and cloud synchronization to support the goal of promoting walking-based physical activity among university students.

Concepts and Technology Used

The STEP-UP application used concepts and technologies related to mobile application development, step tracking, augmented reality, location-based services, gamification, cloud services, and local data handling. These were selected because the application was designed to promote walking-based physical activity through an interactive Android mobile platform.

Mobile application development was one of the main concepts used in the project. STEP-UP was developed for supported Android smartphones because the system requires access to mobile device features such as sensors, camera, location services, internet connection, and local storage. In the application, these device features support step tracking, augmented reality walking mode, campus-based map visualization, and online synchronization.

Automatic step tracking was used to record the walking activity of the user. STEP-UP uses the supported hardware step counter of the Android device as the main basis for counting steps. Android documentation describes the step counter as a motion sensor that reports the number of steps taken by the user while the sensor is activated (Android Developers, 2026). In the system, the recorded steps are used to update progress, complete goals and missions, provide rewards, and support leaderboard ranking.

BMI calculation was included as part of the health guidance feature of the application. After registration or login, the user provides height and weight information, which are used by the system to calculate Body Mass Index. The BMI result is used only as a general wellness reference and as the basis for assigning step-related missions. The mission values are treated as application-based design decisions and are intended for general physical activity support, not medical prescription.

Gamification was applied to make walking more engaging for student users. In STEP-UP, gamification includes daily goals, weekly missions, points, rewards, leaderboard ranking, avatar customization, and progress monitoring. These features were included to make the application more interactive and to encourage users to participate in walking-based activities.

Augmented reality was used to support the AR walking mode of STEP-UP. ARCore is Google's platform for creating augmented reality experiences, and it uses the phone's camera and motion tracking to connect virtual content with the real world (Google for Developers, 2024). In the application, the AR walking mode allows users to experience physical activity with digital elements displayed through a supported Android device.

Location-based technology was used for the campus-based map visualization of STEP-UP. Mapbox Maps SDK for Unity provides tools for building Unity applications using real map data and Mapbox web services through a C#-based API and graphical user interface (Mapbox Docs, 2026). In the system, Mapbox supports map viewing and location-based interaction within the university setting.

Firebase was used to support selected online and cloud-based functions of the application. Firebase Authentication provides backend services and SDKs for authenticating users in an application (Firebase, 2026). In STEP-UP, Firebase supports account login, user-related records, progress data, rewards, leaderboard information, and online synchronization.

Local data handling was also used to support selected offline functions. Since internet connection may be unstable in some campus areas, STEP-UP stores selected user progress and application data locally. When an internet connection becomes available, the system can update or synchronize selected records with the online database. This allows the application to remain usable even when connectivity is limited.

Through these concepts and technologies, STEP-UP was developed as an Android-based mobile application that combines step tracking, BMI-based mission assignment, gamification, augmented reality, campus-based map visualization, Firebase-supported records, and local storage. These components work together to support the application's purpose of promoting walking-based physical activity among university students.

Software

The development of STEP-UP required software tools that supported Android mobile application development, augmented reality, map integration, database management, local data handling, website development, and project collaboration. These tools were selected based on the functions needed by the application and their compatibility with the system design.

Unity was used as the main development environment for the STEP-UP mobile application. It was used to create the application scenes, user interface, avatar system, gamified features, augmented reality walking mode, and other interactive components of the system. Unity also supported the development and building of the application for Android devices through its Android build process and Android Build Support module (Unity Documentation, 2026).

C# was used as the main programming language of the application. It was used to create the scripts and logic that control the functions of STEP-UP, including user interaction, BMI calculation, BMI-based mission assignment, step tracking processing, progress updates, rewards, leaderboard ranking, avatar customization, and data handling. Since Unity uses C# scripting for application behavior, it became the primary language used in implementing the mobile application.

Android development tools were used to build, test, and deploy STEP-UP on supported Android smartphones. These tools supported the application’s connection to Android device features such as sensors, camera, location services, internet connectivity, and storage. The application uses Android-supported device capabilities for automatic step tracking, AR walking mode, and movement-related functions.

ARCore and AR Foundation were used to support the augmented reality feature of STEP-UP. These tools allowed the application to use AR-supported functions on compatible Android devices. In the system, they were used for the AR walking mode, where digital elements are displayed through the device camera and connected with the user’s real-world environment.

Mapbox was used to support the campus-based map visualization of the application. It helped provide the map-related component of STEP-UP, allowing the system to present location-based information connected to the university setting. This feature supports the application’s goal of linking walking activity with the actual campus environment.

Firebase was used to support selected backend and cloud-based functions of STEP-UP. It was used for account-related processes, user records, progress data, rewards, leaderboard information, and online synchronization. Through Firebase, the application can update selected records when an internet connection is available.

Local data storage was used to support selected offline functions of the application. This allows STEP-UP to keep important user progress and application data on the device when internet connection is unstable or temporarily unavailable. Once the connection becomes available, the system can update or synchronize selected records with the online database.

React and Vite were used in the development of the STEP-UP companion website. The website serves as a supporting platform for presenting project information and system-related content. It is separate from the main mobile application but supports the presentation and documentation of the capstone project.

GitHub was used for source code management and collaboration. It allowed the development team to store project files, manage updates, track changes, and organize the progress of the mobile application and website. This helped the researchers maintain the development files throughout the project.

These software tools supported the implementation of the major components of STEP-UP. Through Unity, C#, Android tools, ARCore, Mapbox, Firebase, local storage, React, Vite, and GitHub, the researchers were able to develop the Android mobile application, AR walking mode, campus map feature, gamification system, user records, offline support, online synchronization, and companion website.

Hardware

The STEP-UP application requires supported Android smartphones because the system uses device-based features for step tracking, augmented reality, camera access, location-based services, and internet connectivity. The hardware specifications were identified to ensure that the main functions of the application can run properly during testing and evaluation.

The minimum specifications indicate the lowest device requirements needed to run the major features of STEP-UP. The recommended specifications indicate the preferred device requirements for smoother performance, especially when using augmented reality, campus-based map visualization, and background activity monitoring. Since the application includes AR walking mode, the device should be included in the official ARCore supported devices list. Google explains that ARCore-supported devices are certified based on factors such as camera quality, motion sensors, hardware design, and performance capability, while Android documentation identifies sensors such as the step counter, accelerometer, and gyroscope as motion sensors that may support movement-related functions (Google for Developers, 2026; Android Developers, 2026).

Table 3.1 presents the minimum and recommended hardware specifications for STEP-UP.

Component	Minimum Specifications	Recommended Specifications
Operating System	Android 8.1 (API 27) or higher	Android 11 (API 30) or higher
Device Compatibility	Android device included in the official ARCore Supported Devices list	ARCore-supported Android device with a modern processor, such as Snapdragon 855 equivalent or newer
Memory	Sufficient memory to run the application and AR-related functions	4GB RAM or higher
Sensors	Hardware Step Counter, Accelerometer, and Gyroscope	Hardware Step Counter, Accelerometer, Gyroscope, and Magnetometer/Compass
Camera	Single rear-facing camera	Rear-facing camera with stable autofocus and AR support

Component	Minimum Specifications	Recommended Specifications
Internet Connection	Active internet connection for Mapbox, Firebase, and initial ARCore services update	Stable internet connection for map loading, Firebase synchronization, leaderboard updates, and cloud-based records
Storage	Sufficient storage for application installation and local data	200MB or more free space, excluding possible map caching

Table 3.1. Minimum and Recommended Hardware Specifications for STEP-UP

The hardware step counter is required because STEP-UP uses the supported Android step counter as the main basis for automatic step tracking. The accelerometer and gyroscope support movement and orientation-related functions, while the magnetometer or compass is recommended to improve orientation-related support on capable devices. The rear-facing camera and ARCore-supported device requirement are necessary for the augmented reality walking mode.

An active internet connection is required for features that depend on online services, such as Firebase authentication, data synchronization, leaderboard updates, and Mapbox-related functions. However, selected application data may still be stored locally to support limited use when internet connectivity is unstable or temporarily unavailable. Device performance may still vary depending on the Android version, sensor availability, ARCore compatibility, battery optimization settings, storage condition, and internet connection.

Network Design

The network design of STEP-UP presents how the mobile application communicates with the internet, cloud-based services, location services, and stored user data. Since STEP-UP is an Android-based mobile application, the client tier consists of the PSAU student or user, the Android smartphone, and the STEP-UP mobile application. Through the mobile device, the user can access the application, send requests, and receive data updates from the system.

The network layer represents the connection between the mobile application and online services. STEP-UP uses Wi-Fi or mobile data to connect to the internet. This connection allows the application to perform login and registration, synchronize user records, update leaderboard information, and retrieve map-related services. The Android device also uses GPS or location services to provide location data needed for the map and augmented reality walking features.

The cloud services tier includes Firebase Authentication, Firebase Database or Cloud Storage, and Mapbox Online Map Services. Firebase Authentication supports user login, registration, and account verification. Firebase Database or Cloud Storage stores and synchronizes user-related data such as user profiles, step records, activity records, rewards, avatar data, and leaderboard records. Mapbox Online Map Services provides map and location-based functions used by the application.

The data tier represents the main records handled by the system. These include user accounts, step records, points and rewards, avatar customization data, leaderboard records, and activity history. These data are stored and updated through the cloud services used by the application. This setup allows STEP-UP to manage user progress and support online features without requiring a dedicated university server.

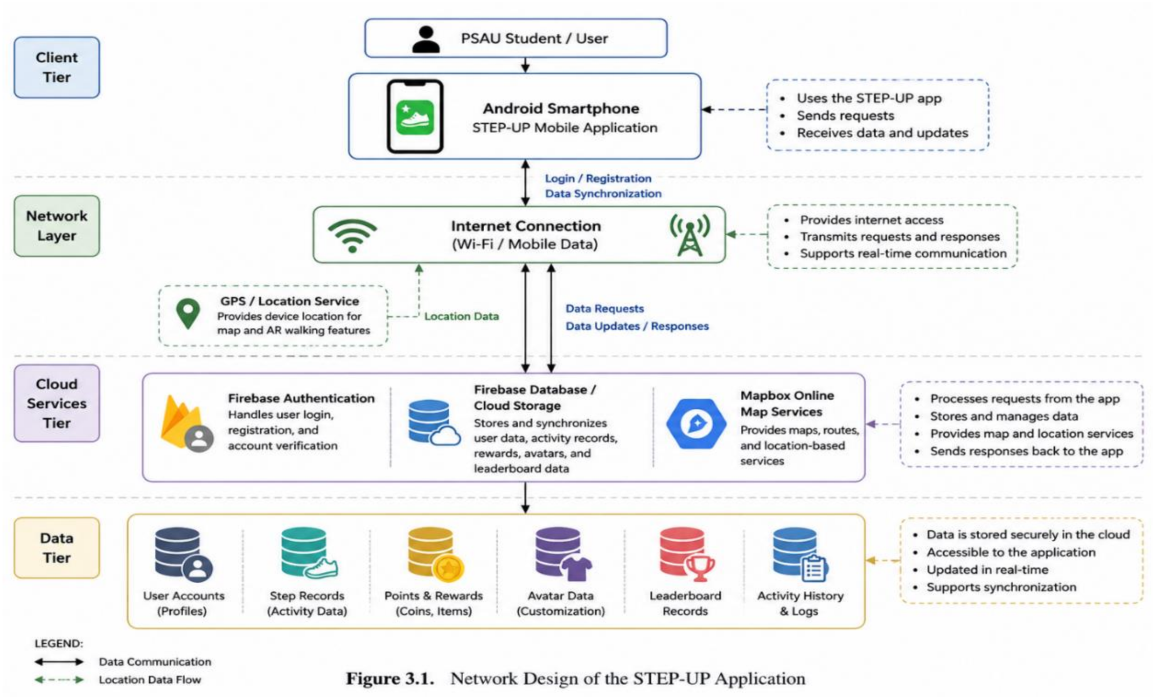

Figure 3.1. Network Design of the STEP-UP Application

Figure 3.1

System Architectural Design

The system architectural design of STEP-UP presents the structure of the application and shows how its main components interact during operation. STEP-UP follows a mobile cloud-based architecture, where the Android smartphone serves as the main user device, the mobile application handles the core functions of the system, and cloud-based services support authentication, data storage, synchronization, and map-related features.

The user layer consists of the PSAU student or user who interacts with the STEP-UP mobile application. Through the Android smartphone, the user can register or log in, view personal information, monitor step count, check activity progress, complete missions, receive rewards, customize an avatar, access the leaderboard, view health guidance, and use the augmented reality walking mode. This layer represents the direct interaction between the user and the application.

The mobile application layer contains the main modules of STEP-UP. These include the user account module, profile and BMI module, automatic step tracking module, activity summary module, gamification and rewards module, avatar customization module, leaderboard module, health guidance module, campus map module, and augmented reality walking mode. These modules work together to support the main purpose of the application, which is to promote physical activity among university students through step-based monitoring and interactive features.

The device services layer represents the built-in hardware and services of the Android smartphone used by the application. The step counter or accelerometer is used to detect and record walking activity. The GPS or location service supports map-based and location-related features, while the camera and ARCore support the augmented reality walking mode. Local storage is also used to keep selected records or settings on the device, especially when the internet connection is unstable or temporarily unavailable.

The cloud services layer includes Firebase Authentication, Firebase Database or Cloud Storage, and Mapbox Online Map Services. Firebase Authentication supports user login and registration, while Firebase Database or Cloud Storage manages user-related records such as profiles, step records, points, rewards, avatar data, and leaderboard information. Mapbox supports the map and location-based features of the application.

The data layer contains the main records handled by the system. These include user accounts, step records, activity history, rewards and points, avatar customization data, and leaderboard records. These data are updated through the interaction between the mobile application, device services, and cloud services. This architecture allows

STEP-UP to provide step tracking, gamified progress, AR walking features, and cloud-based synchronization in one mobile application.

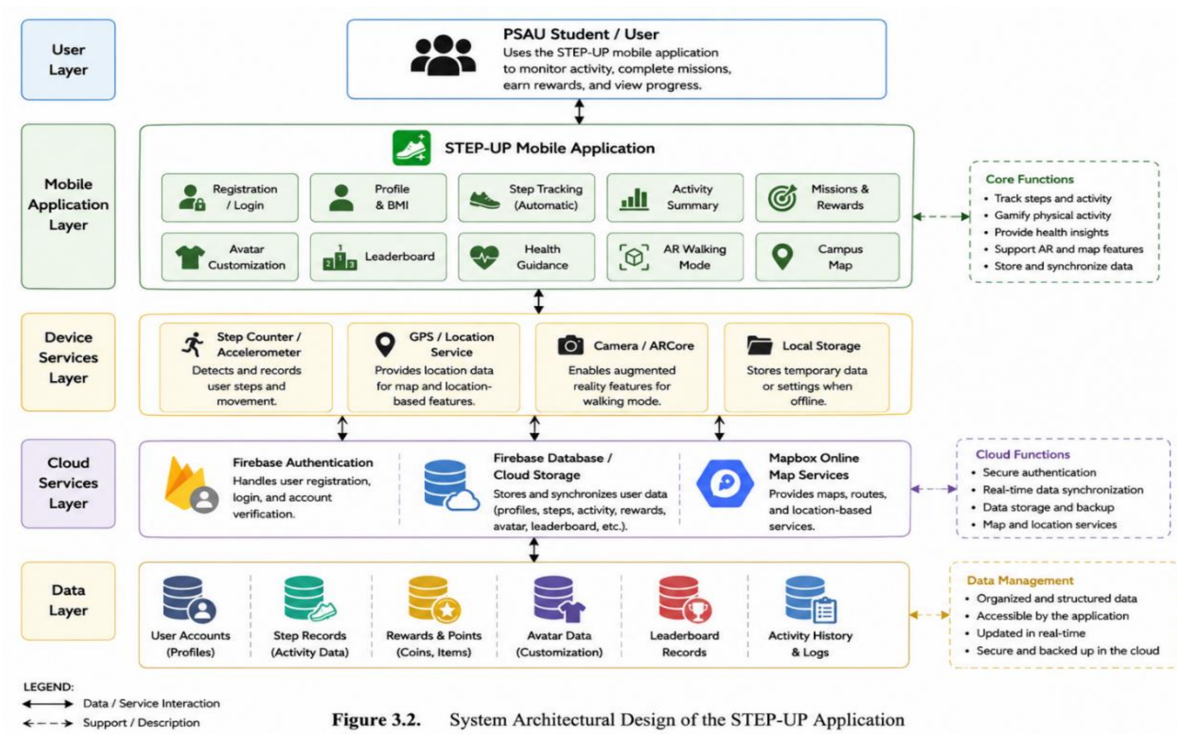

Figure 3.2. System Architectural Design of the STEP-UP Application

Figure 3.2

Features and Functionality of the Proposed System

The expected output of this project is a functional mobile application named **STEP-UP**, designed to promote physical activity among university students through augmented reality, gamification, automatic step tracking, and health guidance features. The application includes several features intended to support user engagement, activity monitoring, progress tracking, and safe participation in walking-based physical activities.

1. **User Registration and Login System -** The application allows users to create an account and securely log in to access personalized step

tracking, progress records, rewards, and other application features. User account data are managed to support continuous access to user records and application progress.

2. **User-Friendly Main Interface -** The application provides an organized and easy-to-use main menu that allows users to access core features such as Play, Customize, Settings, Tutorial, and other system functions. The interface is designed to be simple, intuitive, and suitable for university students.

3. **Local Data Storage System -** STEP-UP includes local data storage to keep relevant user information, step records, earned points, and customization progress available on the device. This feature helps support accessibility and continuity even when the user has limited or no internet connection.

4. **Offline Mode with Online Synchronization -** The application supports offline usage by storing activity data locally. Once internet connectivity is restored, the system synchronizes saved data with Firebase or the cloud database. This allows user progress to be updated even after temporary network interruptions.

5. **Automatic Step Tracking -** The system uses the smartphone's built-in motion sensors to automatically record user steps in real time. This feature allows users to monitor their walking activity without requiring additional devices or manual input.

6. **Activity Summary and Progress Monitoring -** The application provides daily and weekly summaries of step performance, including progress bars and milestone indicators. These summaries help users monitor their physical activity levels and observe their progress over time.

7. **Gamification and Reward System -** STEP-UP converts accumulated steps into in-game points that users can collect and use for rewards. This feature is designed to make physical activity more engaging by connecting walking progress with game-like incentives.

8. **Daily Goals and Weekly Missions -** The application includes milestone-based daily step goals and optional weekly missions. These goals and missions encourage users to participate in consistent walking activities and support gradual improvement in physical activity behavior.

9. **Campus-Based Map Visualization -** The system includes a campus-based map visualization that allows users to view walking-related progress

within the university environment. This feature supports engagement by connecting the user’s physical movement with the campus setting.

10. **Augmented Reality Walking Mode -** STEP-UP includes an augmented reality walking mode that uses the smartphone camera to display the user’s avatar and step progress in real time. This feature is intended to make walking more interactive and immersive. For devices that do not fully support AR functions, the application may provide a fallback display to prevent crashes and maintain usability.

11. **Avatar Customization -** The application allows users to personalize their in-app avatar by unlocking or purchasing outfits and accessories using earned points. This feature supports engagement by giving users a sense of personalization and progression.

12. **Health Guidance Features -** The application provides general health guidance such as posture reminders, warm-up and cooldown tips, and basic fitness information. These features are intended to promote safe and informed physical activity habits.

13. **BMI Calculation, BMI-Based Mission Assignment, Estimated Calories Burned, and Step Recommendation -** The system allows users to input

their height and weight to compute their Body Mass Index (BMI). Based on the user's BMI classification, the application may provide estimated calories burned and step recommendations aligned with general physical activity guidelines.

14. **Vehicle Detection Anti-Cheat Mechanism -** The application includes a speed-threshold mechanism intended to reduce inaccurate step recording during vehicle movement. When movement speed exceeds the set threshold, the system may pause or limit step counting to help maintain the accuracy of recorded activity.

15. **Safety Reminders -** STEP-UP provides safety reminders to encourage users to remain aware of their surroundings while using the application. These reminders help promote responsible use of the application, especially during walking activities and augmented reality interactions.

These features support the main purpose of STEP-UP, which is to encourage physical activity among university students through a combination of step tracking, gamification, augmented reality, progress monitoring, and health-related guidance. The features also provide the basis for evaluating the application in terms of functionality, user engagement, motivation, and software quality.

Project Management

Project management was an important part of the development of STEP-UP because the application required different features to be planned, developed, tested, and improved throughout the project. The system included several components such as user registration and login, EULA flow, BMI calculation, BMI-based mission assignment, step tracking, gamification, avatar customization, Firebase synchronization, Mapbox integration, augmented reality walking mode, offline support, and anti-cheat mechanisms. Because of these requirements, the researchers used an Agile development approach to manage the development process.

Agile Development Methodology

The Agile development methodology was used in the development of STEP-UP to allow the researchers to work on the system through phases of planning, designing, developing, testing, deployment preparation, and review. Agile was appropriate for the project because the application required continuous improvements based on testing results, technical issues, and feature adjustments. Instead of completing the entire system in one fixed process, the researchers improved the application through repeated development and testing activities.

Through Agile, the development team was able to divide the project into manageable tasks. Each part of the application was developed and reviewed based on its function in the system. Features such as the login process, BMI survey, step tracking, missions, rewards, leaderboard, avatar customization, map view, AR walking mode, and Firebase synchronization were developed and improved as the project progressed. This approach helped the team identify problems early and apply corrections before final testing.

The Agile process also allowed the researchers to respond to changes during development. For example, the avatar system was revised to avoid rigging and display problems, the step tracking system was improved using Android-supported step tracking features, and the anti-cheat mechanism was added to help reduce inaccurate step records caused by shaking or vehicle movement. The team also improved the offline mode so selected progress data could be stored locally and synchronized when an internet connection became available.

Agile Development Phases

The development of STEP-UP followed six Agile phases: requirements and planning, design, development, testing, deployment preparation, and review and retrospective. Each phase included specific activities that contributed to the completion and improvement of the system.

Phase 1: Requirements and Planning.

During this phase, the researchers identified the main requirements of the STEP-UP application. The team planned the user onboarding flow, which included login, EULA, BMI survey, and access to the main menu. The BMI-based mission system was also identified as one of the important features of the application. The team planned that the user's BMI result would be used as the basis for assigning step-related missions. The researchers also planned the use of Firebase for persistent user data and Mapbox for campus-based map visualization.

Phase 2: Design.

During the design phase, the researchers prepared the user interface and user experience of the application. The interface was designed to support different Android screen sizes through mobile layout adjustments. The avatar system was also redesigned from a complex body-part system into a lighter full-body mesh-swapping system to reduce rigging and display issues. In addition, the team designed a dual-view structure that allowed the application to switch between the 2D Mapbox view and the 3D augmented reality camera view.

Phase 3: Development.

During the development phase, the researchers implemented the major functions of the application. The team developed the step tracking system, including the StepManager script, to connect the application with the Android device's step tracking capability. Mapbox was integrated to support location-based map visualization, while the augmented reality environment was developed to display the user's avatar in an interactive AR mode. Firebase Realtime Database was also connected to the Unity application to support online synchronization of step counts, unlocked avatar items, progress records, and leaderboard rankings.

Phase 4: Testing.

During the testing phase, the researchers tested the application on physical Android devices. The team checked the step tracking feature, AR mode, Mapbox performance, avatar display, leaderboard updates, and other major functions of the system. Several issues were found and corrected, including avatar display problems, map performance concerns, and step tracking inaccuracies. The team also tested the anti-cheat mechanism after identifying that phone movement and vehicle vibrations could affect step records. To address this, a GPS speed-based checking mechanism was added to help limit step recording when the detected movement exceeded expected walking conditions.

Phase 5: Deployment Preparation.

During this phase, the researchers prepared the application for testing and deployment. The team improved permission handling for features that required activity recognition, camera access, location services, and background activity support. The application was also adjusted to improve touch compatibility on supported Android devices through the updated Unity input system. The React and Vite companion website was also prepared as a supporting platform for presenting project information and system-related content.

Phase 6: Review and Retrospective.

During the review and retrospective phase, the researchers evaluated the completed features and applied final improvements based on testing results. The team polished the user experience, improved offline mode, finalized local caching of selected step and progress data, and prepared synchronization with Firebase when internet connection becomes available. The sensor thresholds and anti-cheat safeguards were also reviewed to support a more stable version of the STEP-UP application.

Agile Phases	**Major Activities**
Planning	Identified system requirements, planned login flow, EULA, BMI survey, BMI-based missions, Firebase data handling, and Mapbox map visualization
Design	Designed mobile UI/UX, adjusted layouts for Android screens, redesigned avatar system, and planned 2D map and 3D AR view switching
Development	Implemented step tracking, BMI-based missions, gamification,

	avatar customization, Mapbox, AR mode, Firebase Realtime Database, rewards, and leaderboard
Testing	Tested the application on Android devices, checked step tracking, AR mode, Mapbox performance, avatar display, Firebase synchronization, and anti-cheat behavior
Deployment Preparation	Improved permissions, Android input compatibility, APK readiness, and companion website preparation
Review and Retrospective	Finalized offline mode, local data caching, Firebase synchronization, sensor thresholds, UI polishing, and final stability improvements

Table 3.2

Development Timeline

The development timeline of STEP-UP presents the sequence of activities completed during the project. The timeline shows how the application was developed from requirements planning up to final review and deployment preparation. The Agile-based timeline includes the planning of the system requirements, design of the interface and application structure, development of the major system modules, testing and debugging, preparation for deployment, and final review of the application.

The Gantt chart of the STEP-UP development timeline is presented in Figure 3.3. It shows the major Agile phases and the corresponding activities completed by the researchers during the development of the system.

Agile Phase	Month 1	Month 2	Month 3	Month 4	Month 5	Month 6	Month 7	Month 8
Requirements and Planning • User flow definition, EULA, BMI survey, mission planning, and planning for Firebase and Mapbox integration.								
Design • Design of UI/UX layout, screen navigation, avatar system, and map and AR view structure.								
Development • Implementation of login, BMI-based missions, step tracking, rewards, leaderboard, and profile features.								
Development • Integration of avatar customization, Firebase synchronization, Mapbox GPS features, AR mode, and companion website.								
Testing • Android device testing for step tracking, Firebase sync, Mapbox, AR mode, interface behavior, and leaderboard updates.								
Deployment Preparation • Permission handling, compatibility checking, APK readiness, and final documentation support.								
Review and Retrospective • Final polishing, bug fixing, system refinement, and stability improvements.								

Figure 3.3